

NANYANG TECHNOLOGICAL UNIVERSITY
SINGAPORE

School of Computer Science and Engineering

SC2104/CE3002: Sensors, Interfacing and Digital Control

Topic 1: Electronics components and circuit interfacing

Assoc Prof Nicholas Vun



Commonly used Electronic Components

In digital circuits and processor systems designs

- electronics components are always needed to support the proper operation of the circuits and systems
  - Resistor
  - Capacitor
  - Inductor
  - Diode
  - Transistor

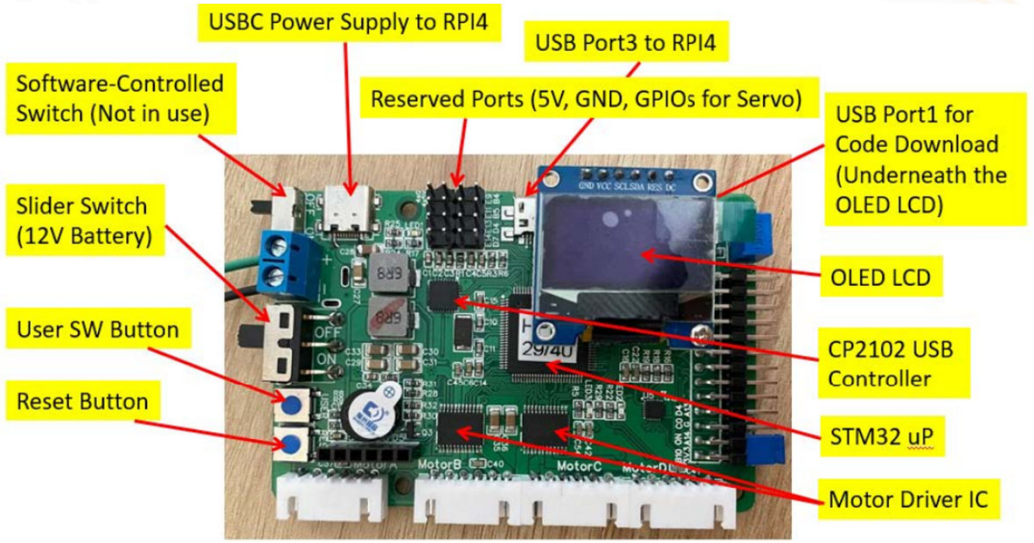
Understanding these components' characteristics and operations will enable us to

- design more reliable systems
- debug problems when something goes wrong

The STM32 Embedded Platform

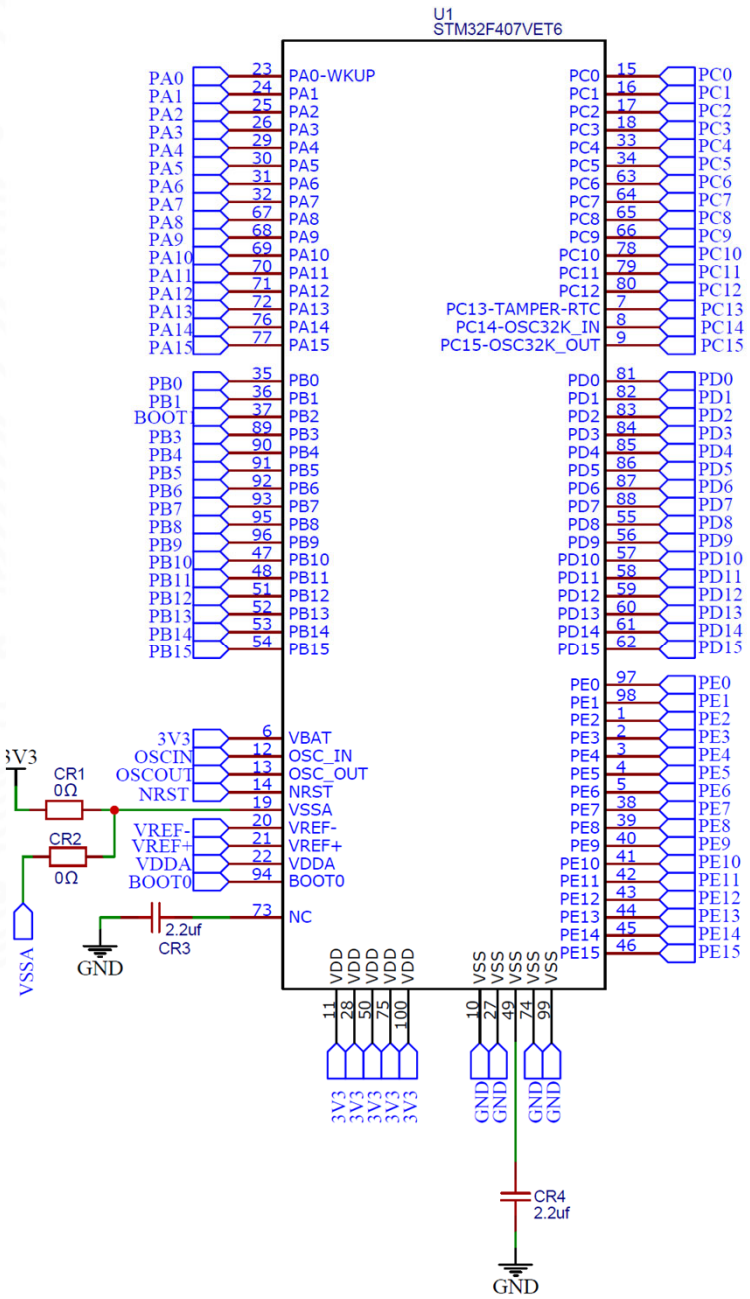
The main embedded platform that we will be using in this course

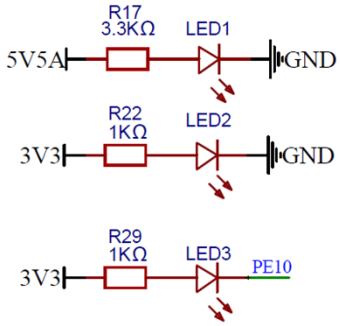
- based on STM32F407VET6 Cortex-M4 processor (the one used in MDP)



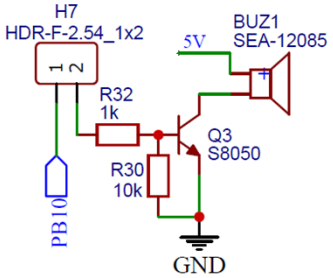
Knowing how to read its schematic is essential to code the programs required to operate the platform

The Schematic

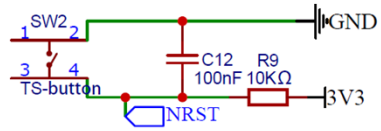




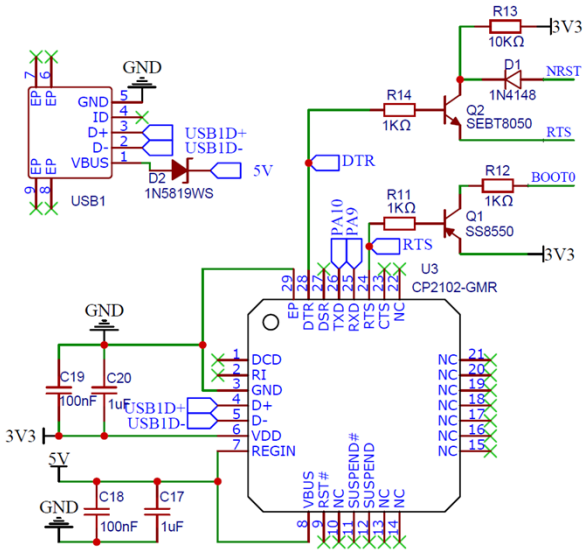
power light



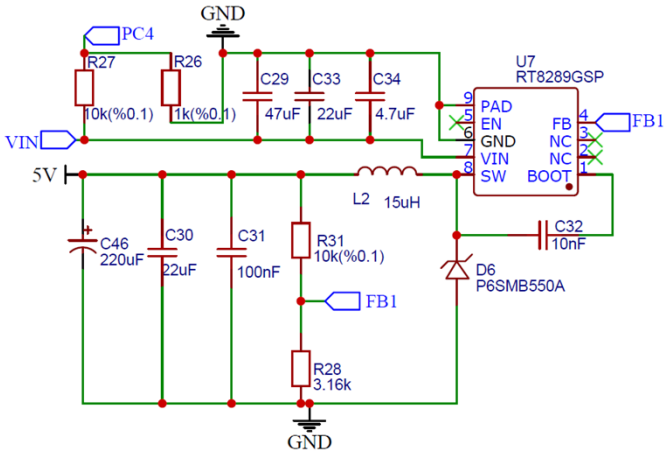
Buzzer



Push Button circuit



Usart1-CP2102



On-board 5V circuit

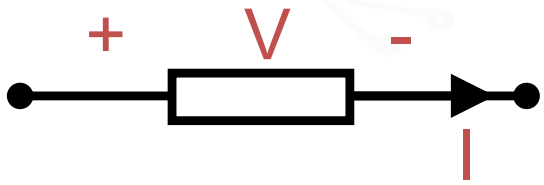
Components V-I Characteristics

Individual electronics component can be described

- by relating its current vs voltage characteristics across its terminals.

For two-terminal components (e.g., R, C, L, Diode)

- we apply voltage across its terminals and monitor current flowing through the component (or vice versa)



For three-terminal devices (e.g. Transistor and Mosfet)

- a bit more complicated

Resistor (and Resistance)



Resistor's characteristics: Resistance Ohm

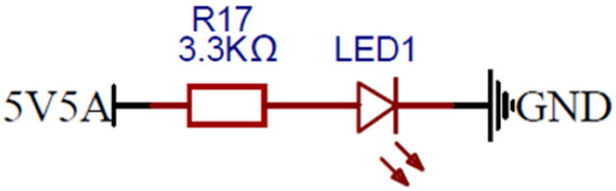
R = Voltage / Current = V / I



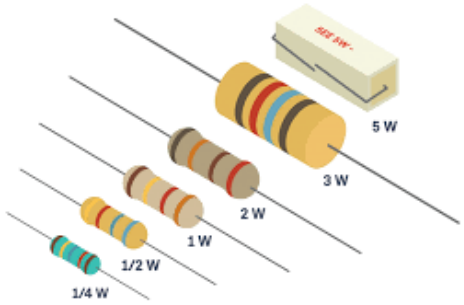
It is frequently used to control the current flowing through the circuit:

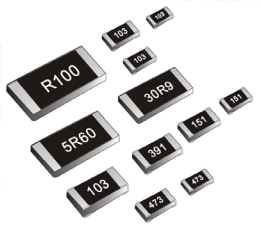
I = V/R

- current flowing through a resistor is proportional to the voltage across its terminal terminals



But watch out for its power dissipation rating: P = V x I

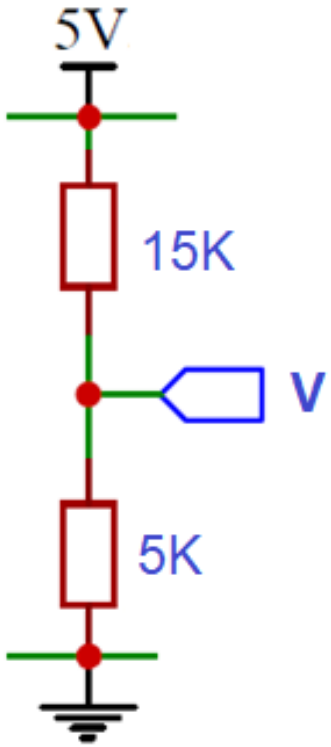




Resistor (cont.)

Resistors are also frequently connected in series to form a voltage divider circuit

- E.g. to establish a 'reference' voltage



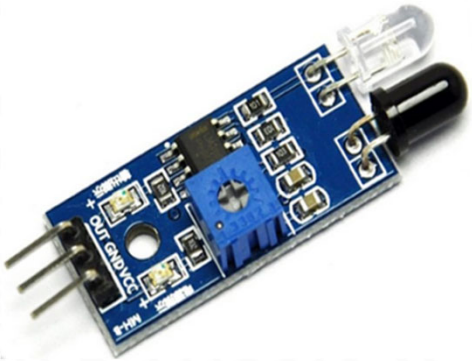
To find the voltage V at the middle of the two resistors:

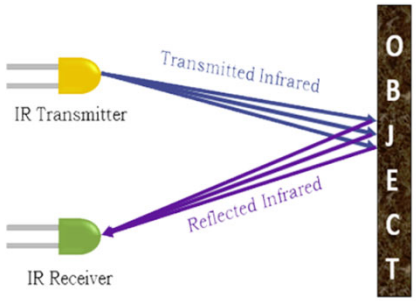
i. calculate the current flowing through the circuit

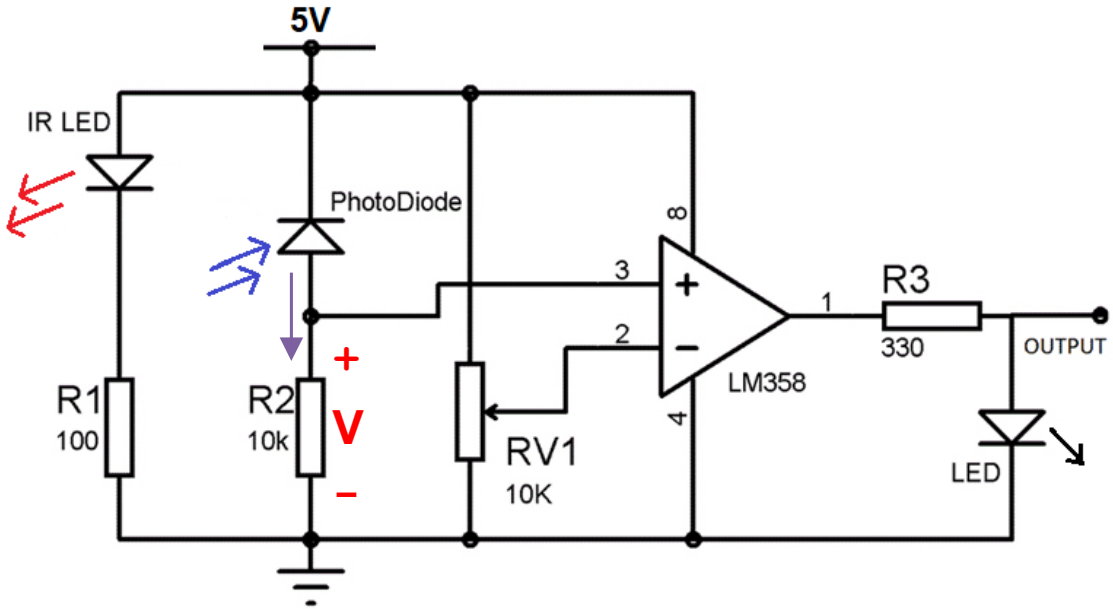
ii. calculate voltage across the individual resistors

Resistor (cont.)

Resistor can also be used to convert current signal to voltage signal (and feed the voltage to the digital circuit)



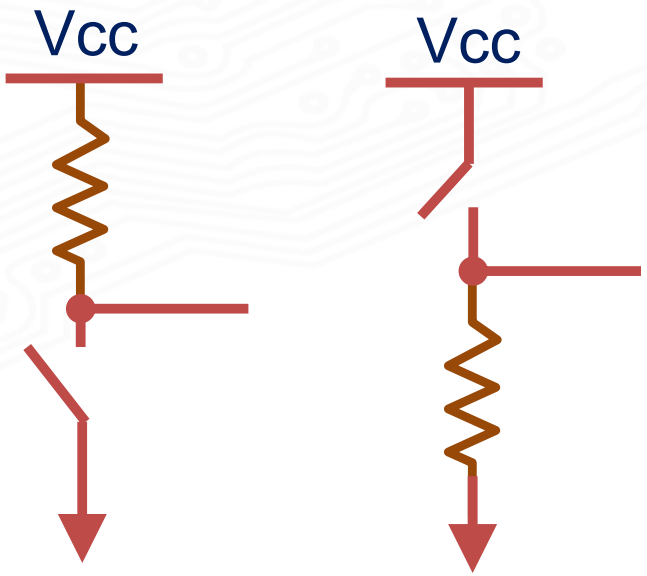


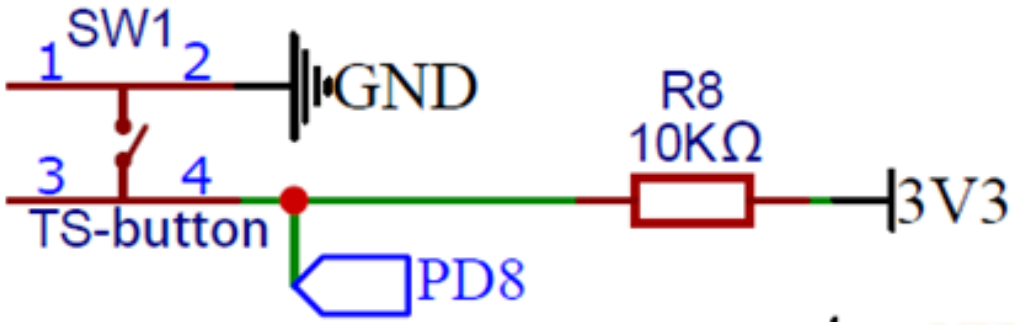


Resistor (cont.)

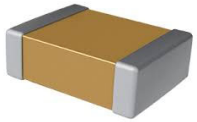
Resistors are commonly used with switches in digital circuits

- used as Pull-Up and Pull-Down resistors
- setup the inactive state of the switch's position





Capacitor (and Capacitance)





Basic capacitor structure

- consists of two closely placed metal plates that are isolated from each other with dielectric materials



Its value, capacitance is expressed in unit of Farads (F)

- but μF and smaller are more useful in practice

Capacitance value of a capacitor depends on

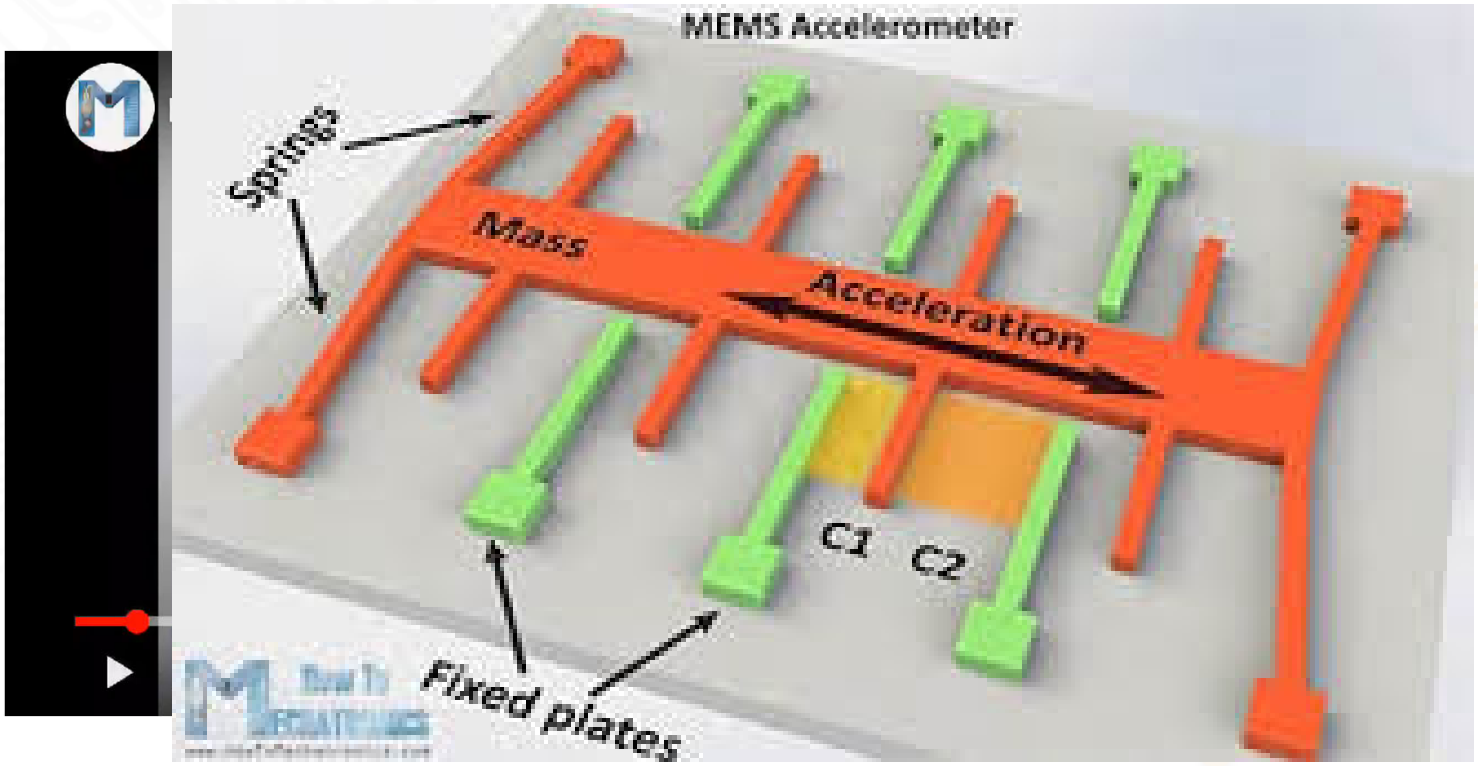
i. the areas of the plates

ii. the distance between the two plates

Capacitor is used in many devices such as touch-screen, microphone and IMU

Capacitance in IMU Device

IMU's Accelerometer and Gyroscope make use of capacitance change to detect movement



Source: https://HowToMechatronics.com

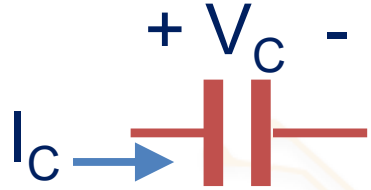
(More about this later)

Capacitor V-I Characteristics

Amount of current flowing through a capacitor C

- proportional to the rate of change of its voltage

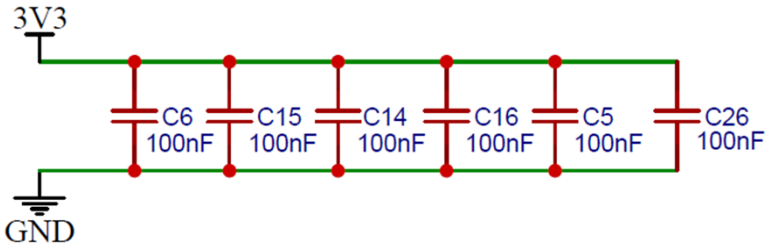
$$I_C = C\, dV_C/dt$$



Practical Implications:

- need large amount of current to change a capacitor's voltage rapidly (i.e. $dV_C/dt \rightarrow \infty$)
- i.e., voltage across the capacitor cannot change instantaneously in practice

Capacitors are hence used extensively in digital circuits to maintain stable voltage in the circuit.



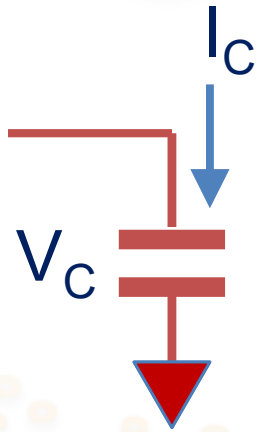
Impedance of a Capacitor (in Frequency Domain)

Consider a capacitor with an applied $V_C = V_P \sin(\omega t)$

$$I_C = C\, dV_C/dt = C\,\omega\, V_P \cos(\omega t)$$

Amplitude of the current: $I_C = C\,\omega\, V_P = V_P / (1/(\omega C))$

- effective 'resistance' = $V_P / I_C = 1/(\omega C)$ with $\omega = 2\pi f$
- i.e., frequency dependent, which is denoted with the term "impedance"
  - i.e., in frequency domain

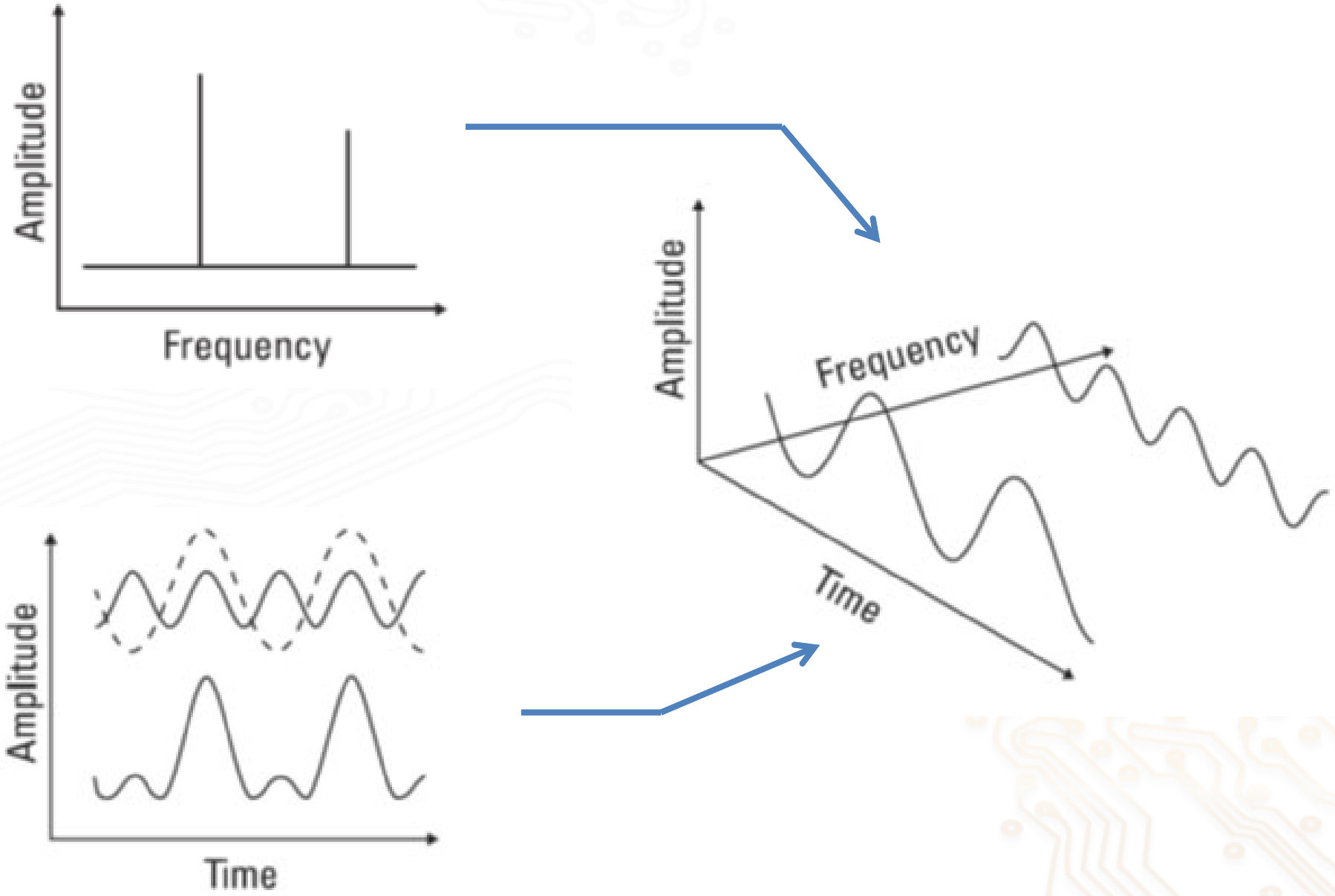


Phase of the current:

- input is sine, output is cosine
- current is leading by 90°
- represented by -j in vector form

Impedance of Capacitor: $Z_C = -j(1/(\omega C)) = 1/(j\omega C)$

Aside: Time Domain vs Frequency Domain



Impedance of a Capacitor (in Frequency Domain)

Impedance of Capacitor:

$$Z_C = -j(1/(\omega C)) = 1/(j\omega C)$$

i. $Z_C \rightarrow \infty$ as $\omega \rightarrow 0$,
   - i.e. behaving like an open circuit at low frequency

ii. $Z_C \rightarrow 0$ when $\omega \rightarrow \infty$
   - i.e. becoming a short circuit at high frequency

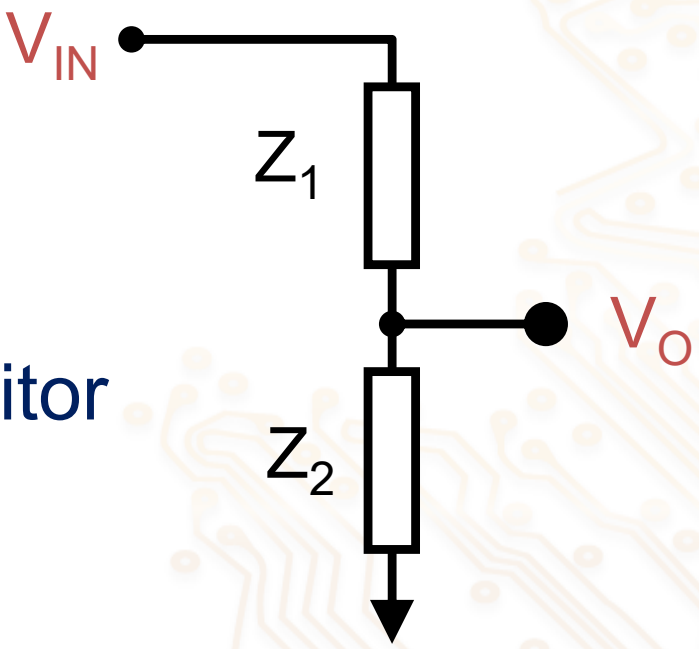
Frequency Dependent Circuit

By combining components such as resistor & capacitor in a circuit

- we can form frequency dependent circuit that are very useful in practice
  - E.g., to reduce/remove noise from the signal before feeding it to the processor

Simplest arrangement:

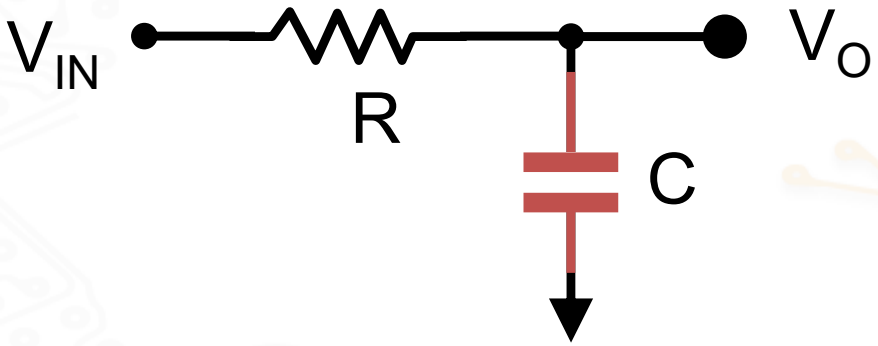
- voltage-divider circuit
- Z_1 is a resistor and Z_2 is a capacitor (and vice-versa)



Low Pass RC Filter

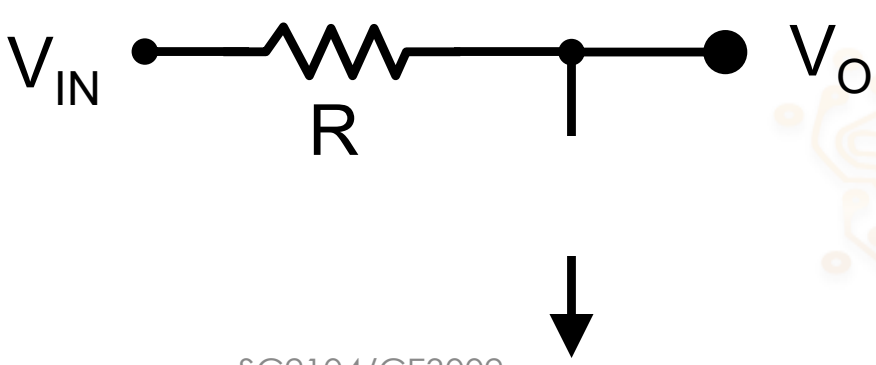
This RC circuit forms a low pass filter

- i.e., only allow low frequency signal to go through



i. At low frequency, C behaves like an open circuit

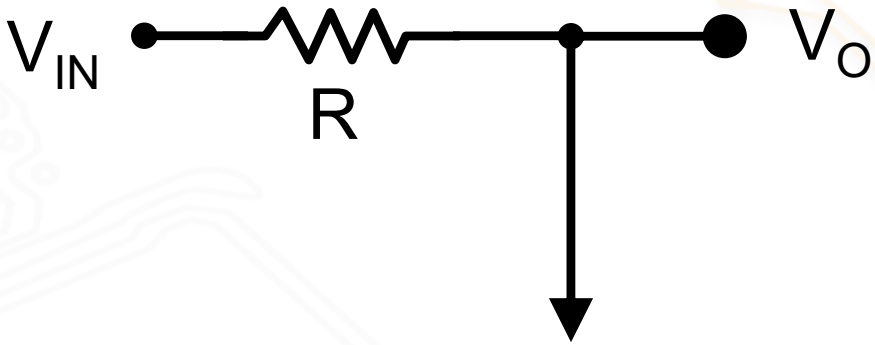
- Hence $V_O = V_{IN}$
- i.e. low frequency signal will pass through the circuit



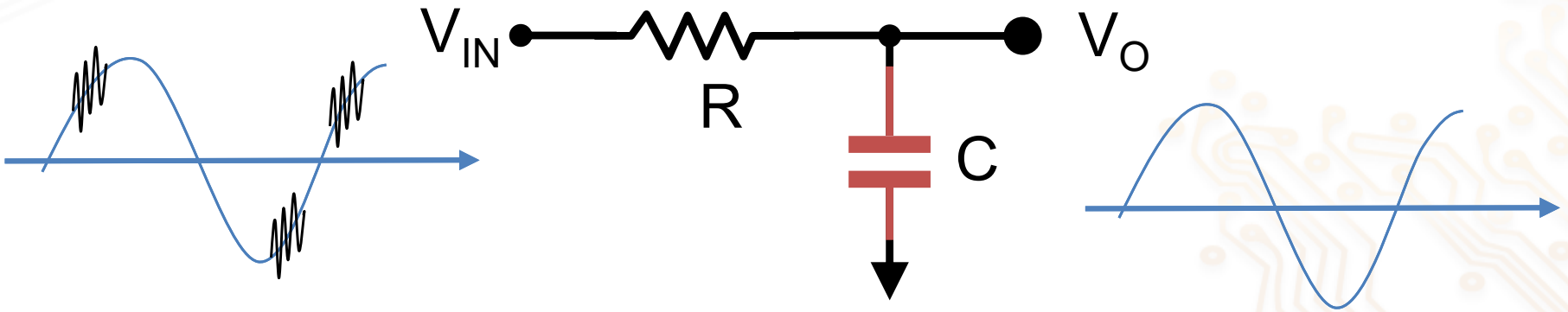
Low Pass RC Filter

ii. At very high frequency, C behaves like a short circuit

- Hence $V_O = 0$
- i.e. high frequency signal will be short circuit to ground.



Example: An ideal case of filtering



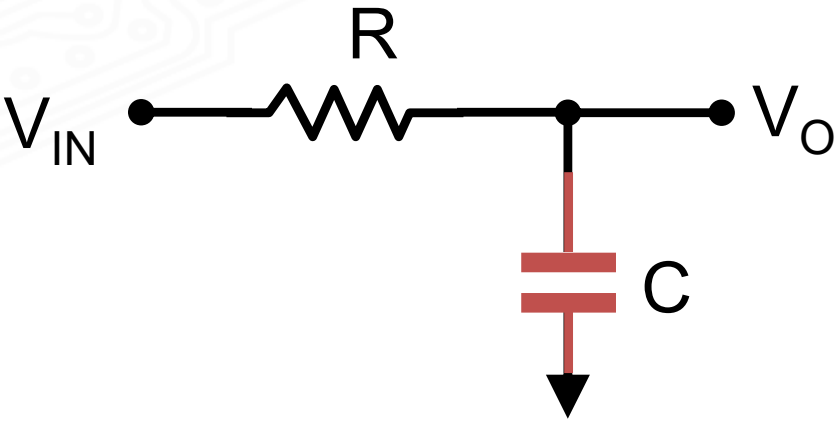
Bandwidth of Filter

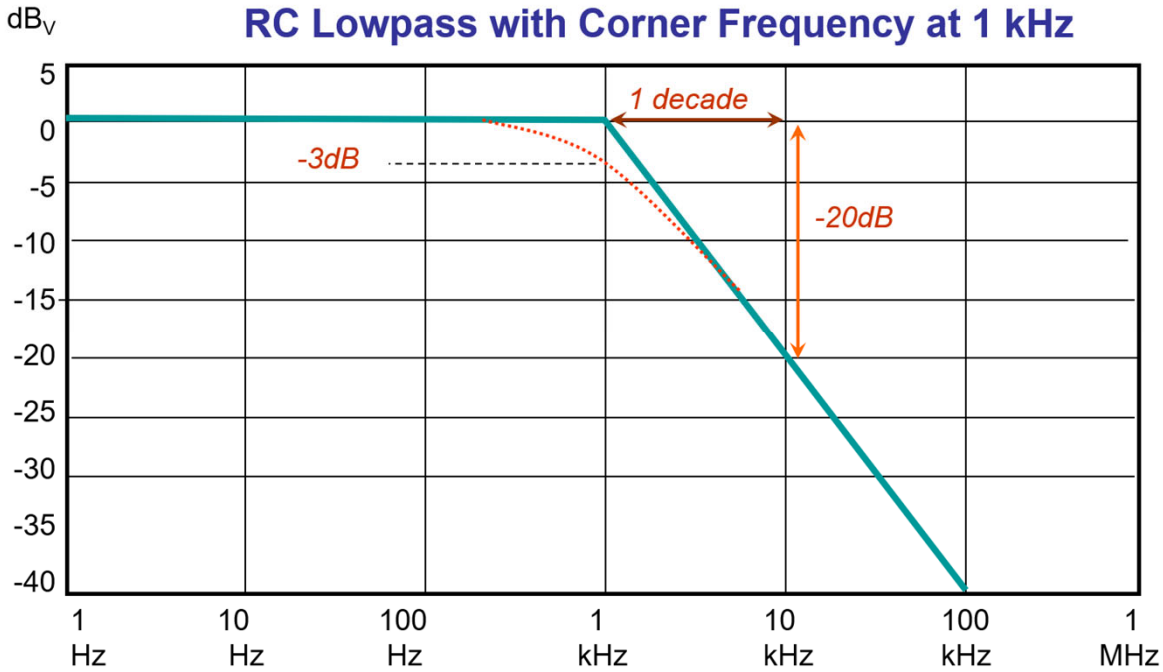
Bandwidth is used to describe the frequency range that the filter will allow the signal to pass to the output

- defined as the frequency where the output power of the signal is reduced to half of its input power

Formula of the RC filter's bandwidth = $1/(2\pi RC)$

- also known as the corner frequency f_C

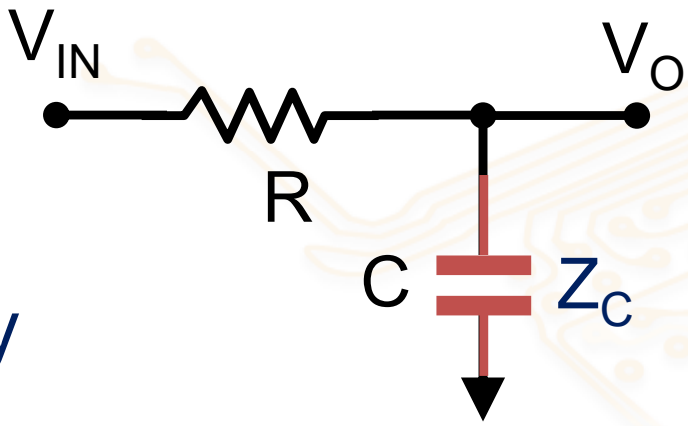




What happens at Corner Frequency?

A more insightful way to understand the operation of the circuit:

- at low frequency: $Z_C = \infty$
- at high frequency: $Z_C = 0$



So as the frequency of the signal

is changed from low to high frequency

- Z_C changes from $\infty \rightarrow 0$
- there will be a particular frequency where $|Z_C| = R$
  - which turn out to be at $f_C = 1/(2\pi RC)$

Hence corner frequency is the point where the impedances of the capacitor equals to the resistor

-3db Corner Frequency

As $Z_C = 1/(j\omega C)$ is 90° out of phase with the R value

- amplitude of V_O can be shown to be 0.707 V_{IN} at the Corner Frequency, i.e., $V_O/V_{IN} = 0.707$

A common way to compare the amplitudes of input and output signal of a circuit

- using the Decibel (dB) convention: 20 Log (V_O/V_{IN})
- 20 Log (0.707) = -3dB

Hence bandwidth is also known as the -3dB corner frequency.

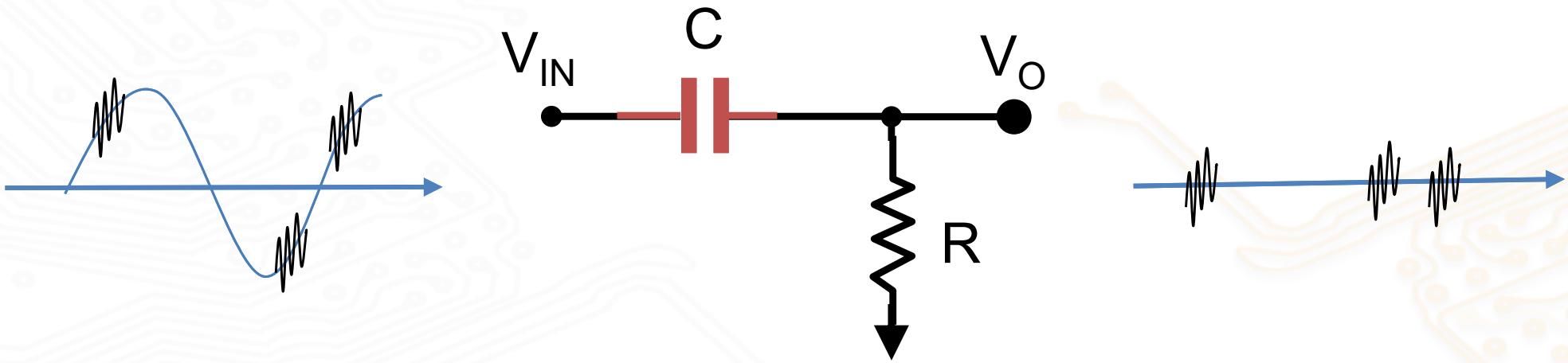
(Note that power is proportional to V^2

- output power at corner frequency = $0.707^2 = 0.5$
- i.e. half of the input power - the bandwidth definition)

High Pass RC Filter

By swapping the positions of the R & C

- the circuit becomes a High Pass filter.



(HP Filter will be discussed in more detail during the tutorial)

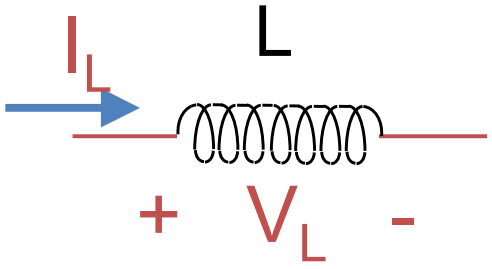
More sophisticated filters can be implemented in software

- i.e. Digital Filters – which will be covered in detail later

Inductor (and Inductance)

Inductor V-I characteristics:

$$V_L = L\, dI_L/dt$$

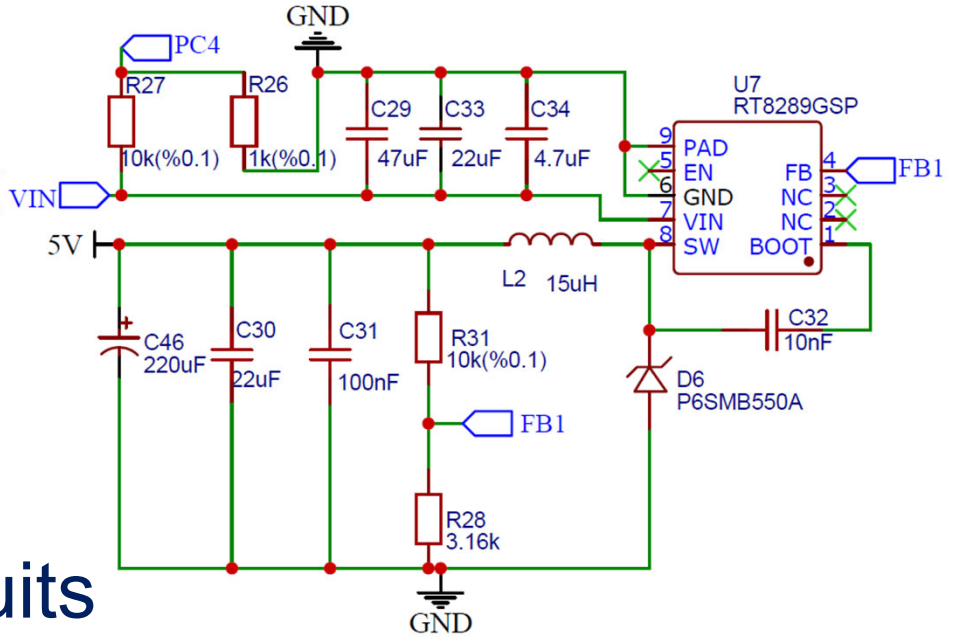


- voltage will be induced when there is a change in the current
- Unit: Henry (but μH in practice)
- Impedance $Z_L = j\omega L$

Inductors are inherently found in many equipment

- E.g., DC motors

Inductors are also used in circuits such as power supply

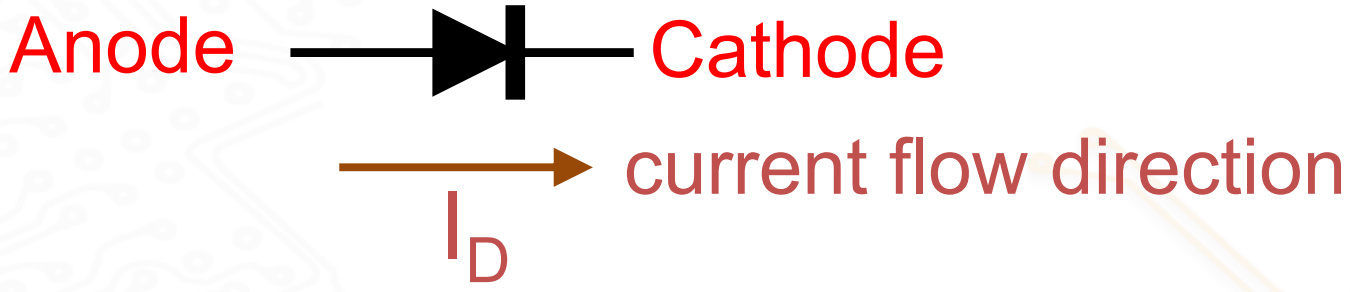


On-board 5V circuit

Diode

Diodes are semiconductor component

- with two terminals: “Anode” and “Cathode”

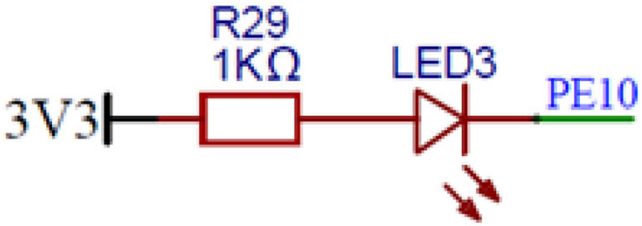


Behaves like a simple on/off switch

- but conducts current in only one direction
- controlled by voltage applied
  - i.e. an 'Electronic' switch

Power is dissipated as heat

- or light energy (i.e., LED)



Ideal Diode

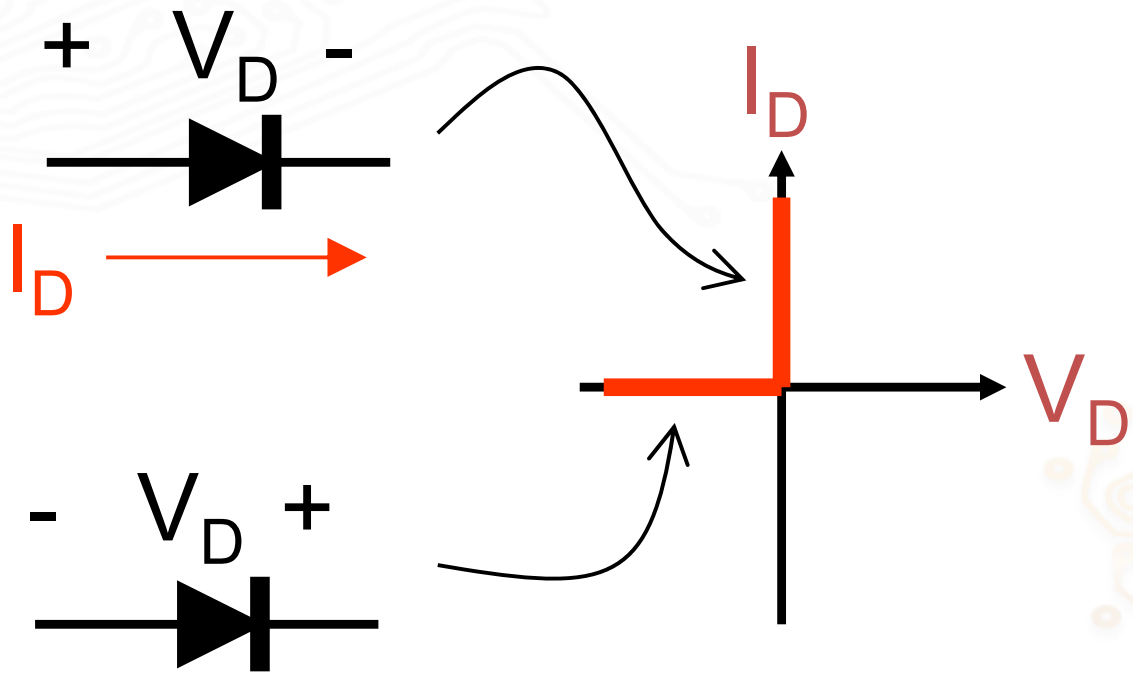
The ideal diode behaves like

- an open circuit in the non-conduction region

  Reversed Biased : Anode is -ve w.r.t. the Cathode

- a short circuit in the conduction region

  Forward Biased : Anode more +ve than Cathode



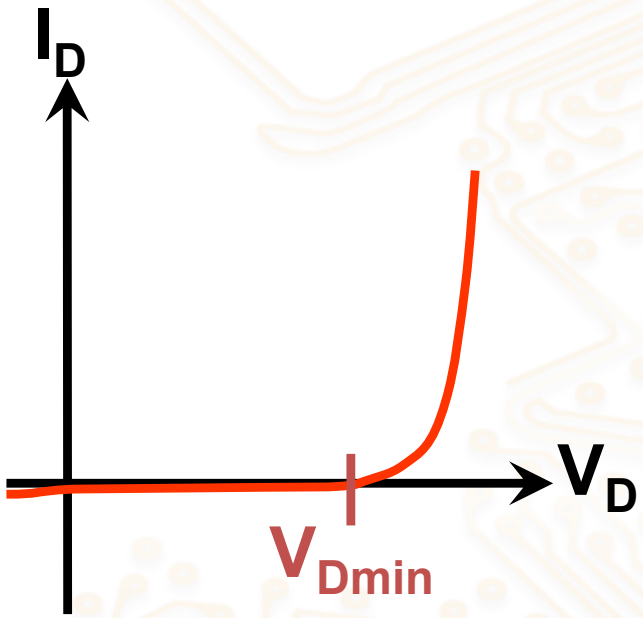
Practical Diodes

Practical diodes have forward voltage drop when current is flowing

- values depends on the types of diode

Typical V_{Dmin} values:

a. Small signal diode = 0.7V
b. Light Emitting Diode = 2V to 3V
c. Infra Red Diode = 1.2V to 1.4V
d. Schottky diode = 0.2V

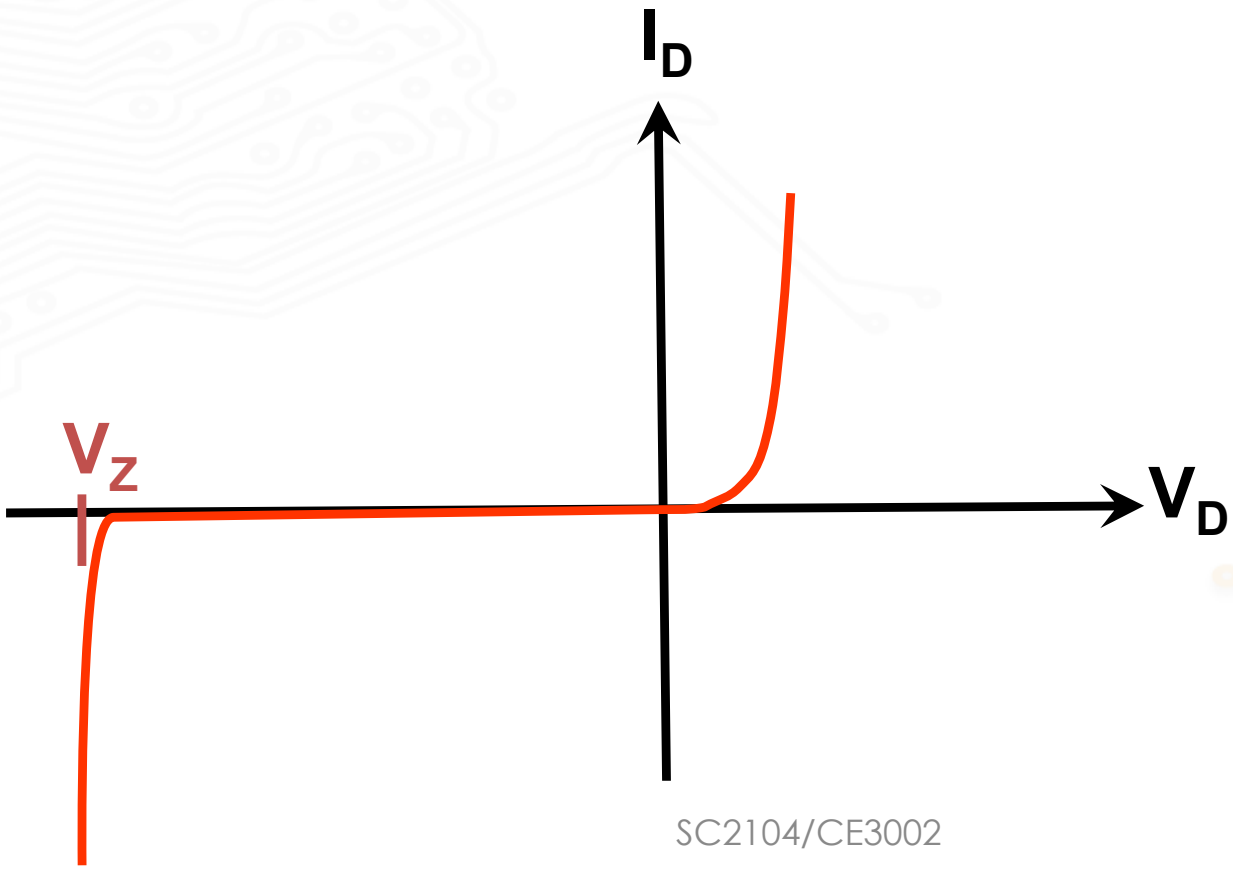


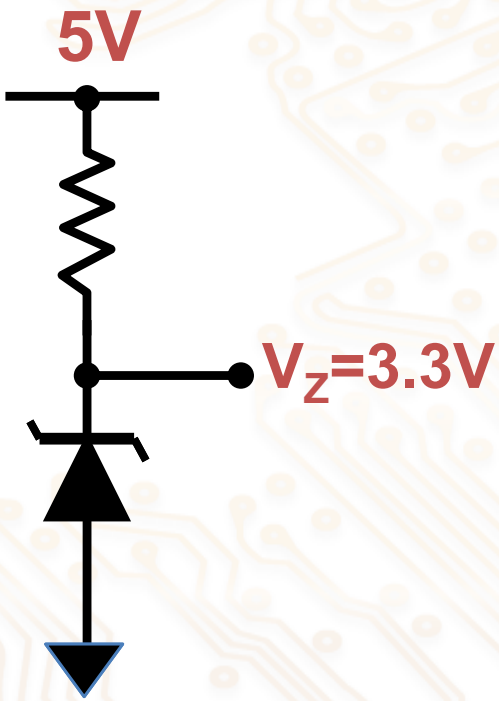
Aside: Zener Diode

A special kind of diode that is purposely connected in the reverse (biased) direction

- such that it 'breaks down' with a constant voltage V_Z

Can be used as a simple voltage regulator circuit

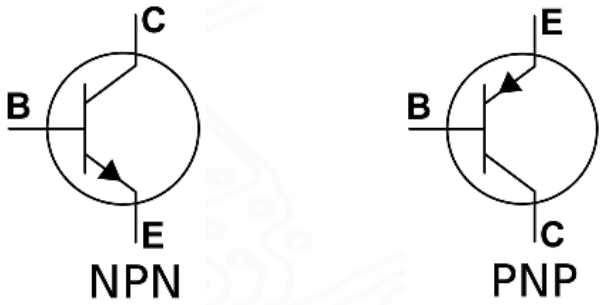




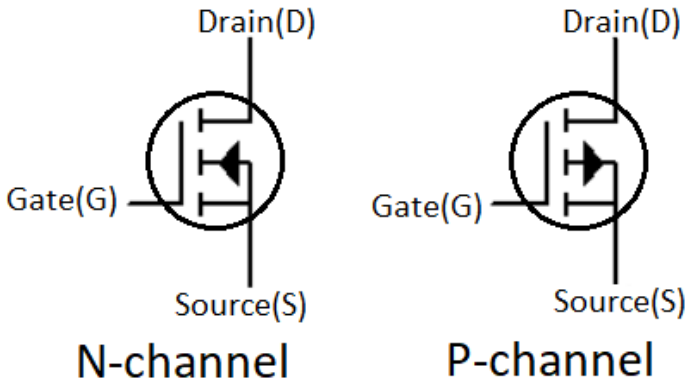
Transistor

There are two main types of transistors used in digital circuits

i. Bipolar Junction Transistor (BJT)



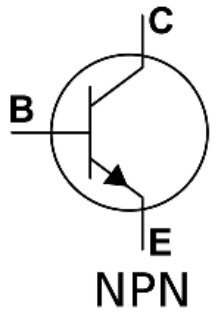
ii. Field Effect Transistor (Mosfet)



BJT Transistor

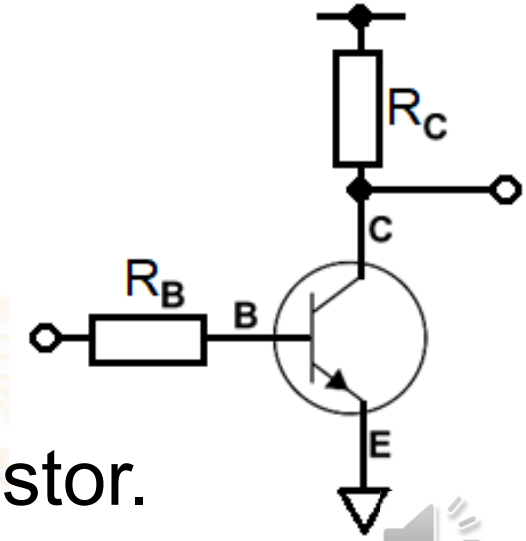
BJT is three terminals current driven device

- input signal **current** is applied to the B (Base) terminal of the transistor
- cause a bigger current (e.g. 10x) to flow from terminal C (Collector) to terminal E (Emitter) - for an NPN BJT



Resistor R_B is hence added to the Base to convert input voltage to current.

Output current can be converted back to voltage if needed by using another resistor.



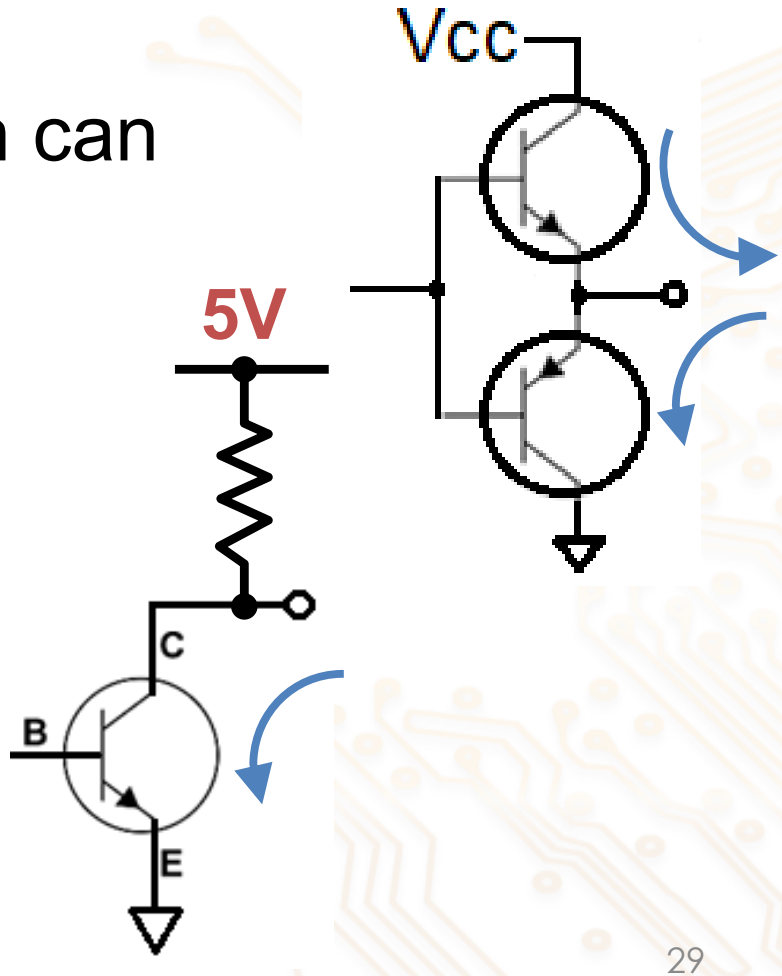
Output Stage Circuit

BJTs are commonly used in the I/O pins of embedded processor as output drive circuit

- need to be properly configured before operation

Typical configuration choices are

1. Push-Pull output stage, which can source and sink load current
   - i.e., output '1' and '0'
   - e.g., to drive a motor
2. Open Collector output stage
   - can only sink current,
     - i.e., logic '0' only
   - a pull-up resistor can be added to support logic '1'



Summary

We cover the main components that are commonly used or encountered in design and implementation of digital systems.

Understanding the operations and proper usages of these components

- enable us to design and implement more robust digital systems
- know how to configure the I/O interfaces of embedded system for proper operation

  (which you will be doing during the laboratory exercise of this course.)